Workshop: One-Way ANOVA and Kruskal-Wallis

your name here

This week, we continue working with the Totally Serious Pop Culture Research Institute™ on research questions in Equestria.

Let's call in the packages and data we need,

```
library(tidyverse)
library(ssstats)
applebuck_recovery       <-       read_csv("https://raw.githubusercontent.com/
samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-2-applebuck_
recovery.csv")
crystal_focus <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/M2-2-crystal_focus.csv")
friendship_learning      <-       read_csv("https://raw.githubusercontent.com/
samanthaseals/STA4173Fa26/refs/heads/main/workshops/data/M2-2-friendship_
learning.csv")
```

Study 1: Crystal Focus

The Totally Serious Pop Culture Research Institute™ has opened an Equestria branch! Twilight Sparkle conducted an experiment to determine whether preparation method and pony type are associated with focus during a crystal-focus task. Researchers recruited 90 ponies: 30 Earth ponies, 30 pegasi, and 30 unicorns. Within each pony type, 15 ponies were randomly assigned to quiet practice and 15 to guided visualization.

After preparing, each pony completed the same task and received a focus score from 0 to 100, with higher scores indicating better focus. The researchers want to determine whether the effect of preparation method on average focus scores depends on pony type.

Dataset name: `crystal_focus`

Variables:

- `pony_id`: unique pony identifier.
- `pony_type`: Earth pony, Pegasus, or Unicorn.
- `preparation`: assigned condition.
- `focus_score`: response described above.

1a. Find the summary statistics to describe this data for each combination of the two explanatory variables.

```
crystal_focus |>
  group_by(pony_type, preparation) |>
  mean_median(focus_score)
```

```
# A tibble: 6 × 5
  pony_type  preparation          variable    mean_sd    median_iqr
  <chr>      <chr>                <chr>       <chr>      <chr>
1 Earth pony Guided visualization focus_score 86.5 (4.9) 85.8 (7.0)
2 Earth pony Quiet practice       focus_score 65.4 (3.9) 65.3 (4.4)
3 Pegasus    Guided visualization focus_score 74.0 (7.1) 73.5 (8.8)
4 Pegasus    Quiet practice       focus_score 75.4 (3.9) 76.2 (5.4)
5 Unicorn    Guided visualization focus_score 64.2 (3.9) 63.9 (5.2)
6 Unicorn    Quiet practice       focus_score 87.3 (5.6) 88.5 (6.2)
```

1b. Construct the appropriate two-way ANOVA table to determine if there is an interaction between preparation and pony type.

```
crystal_focus |> anovas(outcome = focus_score,
                        A = preparation,
                        B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source                 df         SS         MS        F  p-value
──────────────────────────────────────────────────────────────────
preparation             1    29.4694    29.4694    1.171    0.282
pony_type               2    27.7760    13.8880    0.552    0.578
preparation:pony_type   2  7340.4436  3670.2218  145.780  < 0.001
Residuals              84  2114.8200    25.1764
──────────────────────────────────────────────────────────────────


Hypotheses (preparation:pony_type):
H₀: There is no interaction between preparation and pony_type
H₁: There is an interaction between preparation and pony_type
Test statistic: F(2, 84) = 145.78
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

1c. Conduct the appropriate hypothesis test to show that there is an interaction effect.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $F =$
- Degrees of freedom:
- $p =$

Rejection Region

- Reject H_0 if ...

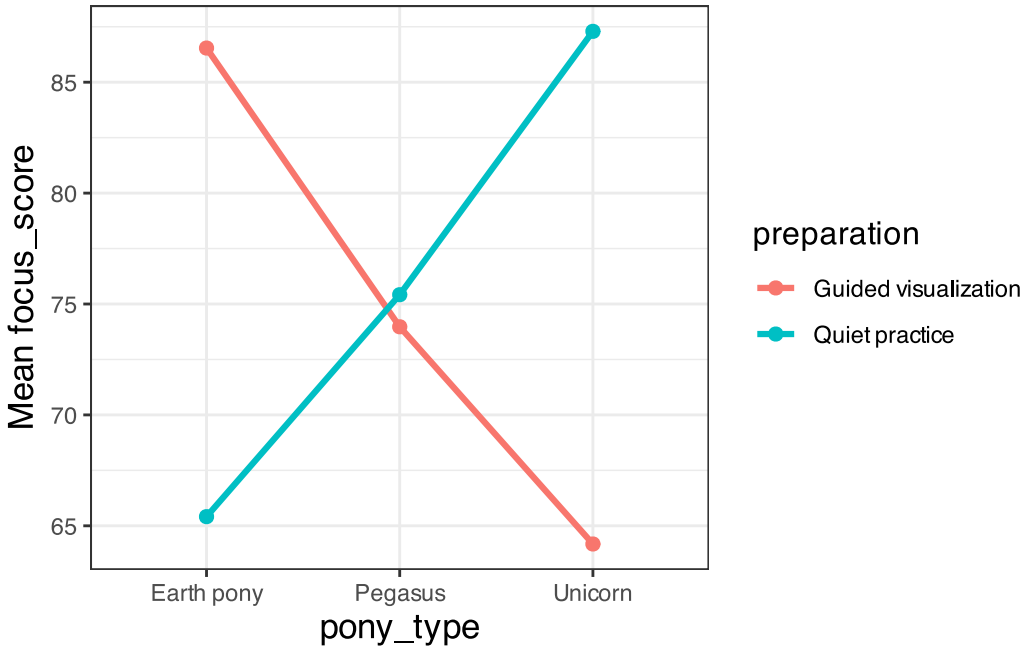
Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

1d. Construct the appropriate posthoc test for a confirmatory study.

1e. Construct a profile plot for this data.

```
crystal_focus |> profile_plot(outcome = focus_score,
                              xaxis = pony_type,
                              lines = preparation)
```



1f. Write a few sentences to describe what's going on with the data using the results of the ANOVA, the posthoc testing, and the profile plot above.

Your answer here.

Study 2: School of Friendship

The Totally Serious Pop Culture Research Institute worked with Starlight Glimmer to compare lesson formats at the School of Friendship. Researchers recruited 90 ponies: 30 Earth ponies, 30 pegasi, and 30 unicorns. Within each pony type, 15 ponies were randomly assigned to an individual lesson and 15 to a cooperative lesson.

After the lesson, each pony completed the same assessment scored from 0 to 100, with higher scores indicating greater learning. The researchers want to determine whether the effect of lesson format on average learning scores depends on pony type.

Dataset name: `friendship_learning`

Variables:

- `pony_id`: unique pony identifier.
- `pony_type`: Earth pony, Pegasus, or Unicorn.
- `lesson_format`: assigned condition.
- `learning_score`: response described above.

2a. Find the summary statistics to describe this data for each combination of the two explanatory variables.

```
friendship_learning |>
  group_by(lesson_format, pony_type) |>
  mean_median(learning_score)
```

```
# A tibble: 6 × 5
  lesson_format pony_type  variable       mean_sd    median_iqr
  <chr>         <chr>      <chr>          <chr>      <chr>
1 Cooperative   Earth pony learning_score 69.1 (5.2) 67.8 (4.8)
2 Cooperative   Pegasus    learning_score 80.7 (5.3) 79.8 (6.2)
3 Cooperative   Unicorn    learning_score 89.0 (4.6) 88.9 (5.7)
4 Individual    Earth pony learning_score 57.5 (5.4) 57.4 (8.0)
5 Individual    Pegasus    learning_score 70.6 (4.1) 70.8 (6.8)
6 Individual    Unicorn    learning_score 81.9 (3.3) 82.9 (5.1)
```

2b. Construct the appropriate two-way ANOVA table to determine if there is an interaction between preparation and pony type.

```
friendship_learning |> anovas(outcome = learning_score,
                              A = lesson_format,
                              B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source                  df         SS         MS        F  p-value
```

```
──────────────────────────────────────────────────────────────────────
lesson_format             1  2072.6401  2072.6401   93.132  < 0.001
pony_type                 2  7390.4047  3695.2023  166.040  < 0.001
lesson_format:pony_type   2    75.1442    37.5721    1.688    0.191
Residuals                84  1869.4120    22.2549
──────────────────────────────────────────────────────────────────────


Hypotheses (lesson_format:pony_type):
H₀: There is no interaction between lesson_format and pony_type
H₁: There is an interaction between lesson_format and pony_type
Test statistic: F(2, 84) = 1.688
p-value: 0.191
Conclusion: Fail to reject the null hypothesis (p = 0.191 ≥ α = 0.050)
```

2c. Conduct the appropriate hypothesis test to show that there is NOT an interaction effect.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $F =$
- Degrees of freedom:
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

2d. Reconstruct the two-way ANOVA table without the interaction term.

```
friendship_learning |> anovas(outcome = learning_score,
                              A = lesson_format,
                              B = pony_type,
                              interaction = FALSE)
```

```
Two-way ANOVA (no interaction)

Source         df         SS         MS        F  p-value
──────────────────────────────────────────────────────────
lesson_format   1  2072.6401  2072.6401   91.665  < 0.001
```

```
pony_type       2  7390.4047  3695.2023  163.424  < 0.001
Residuals      86  1944.5562    22.6111
──────────────────────────────────────────────────────────


Hypotheses (lesson_format):
H₀: All means across levels of lesson_format are equal
H₁: At least one mean across levels of lesson_format differs
Test statistic: F(1, 86) = 91.665
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)

Hypotheses (pony_type):
H₀: All means across levels of pony_type are equal
H₁: At least one mean across levels of pony_type differs
Test statistic: F(2, 86) = 163.424
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

2e. Conduct the appropriate hypothesis test to determine if there is a relationship between learning score and pony type.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $F =$
- Degrees of freedom:
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

2f. Conduct the appropriate hypothesis test to determine if there is a relationship between learning score and lesson format.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $F =$
- Degrees of freedom:
- $p =$

Rejection Region

- Reject H_0 if ...

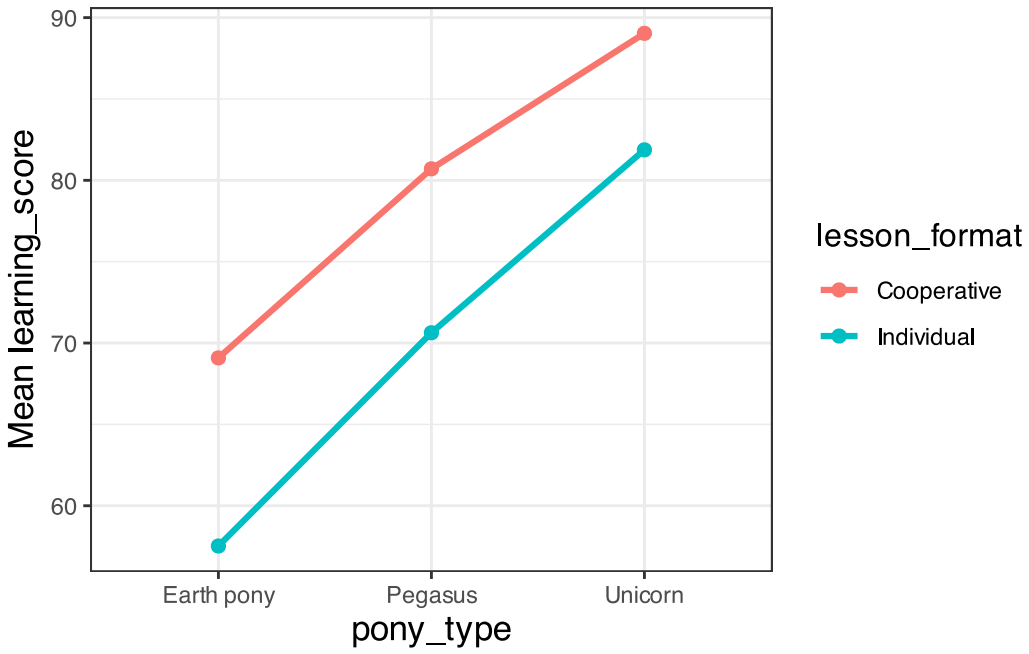
Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

2g. Construct the appropriate posthoc tests for an exploratory study.

2h. Construct a profile plot for this data.

```
friendship_learning |> profile_plot(outcome = learning_score,
                                    lines = lesson_format,
                                    xaxis = pony_type)
```



2i. Write a few sentences to describe what's going on with the data using the results of the ANOVA, the posthoc testing, and the profile plot above.

Your answer here.

Study 3: Sweet Apple Acres

The Totally Serious Pop Culture Research Institute partnered with Applejack to investigate recovery after applebucking. Researchers recruited 90 ponies: 30 Earth ponies, 30 pegasi, and 30 unicorns. After a standardized applebucking session, 15 ponies within each pony type were randomly assigned to rest and 15 to stretching.

Researchers recorded the number of minutes until each pony returned to its pre-session heart rate, with lower values indicating faster recovery. The researchers want to determine whether the effect of recovery routine on average recovery time depends on pony type.

Dataset name: `applebuck_recovery`

Variables:

- `pony_id`: unique pony identifier.
- `pony_type`: Earth pony, Pegasus, or Unicorn.
- `recovery_routine`: assigned condition.
- `recovery_minutes`: response described above.

2a. Find the summary statistics to describe this data for each combination of the two explanatory variables.

```
applebuck_recovery |>
  group_by(recovery_routine, pony_type) |>
  mean_median(recovery_minutes)
```

```
# A tibble: 6 × 5
  recovery_routine pony_type  variable         mean_sd    median_iqr
  <chr>            <chr>      <chr>            <chr>      <chr>
1 Rest             Earth pony recovery_minutes 63.3 (4.8) 62.2 (4.6)
2 Rest             Pegasus    recovery_minutes 68.5 (5.5) 67.5 (5.1)
3 Rest             Unicorn    recovery_minutes 61.0 (5.0) 60.6 (3.9)
4 Stretching       Earth pony recovery_minutes 59.1 (3.9) 57.6 (5.3)
5 Stretching       Pegasus    recovery_minutes 50.5 (4.5) 50.5 (4.6)
6 Stretching       Unicorn    recovery_minutes 39.7 (4.3) 39.0 (7.5)
```

2b. Construct the appropriate two-way ANOVA table to determine if there is an interaction between recovery routine and pony type.

```
applebuck_recovery |> anovas(outcome = recovery_minutes,
                             A = recovery_routine,
                             B = pony_type)
```

```
Two-way ANOVA (with interaction)

Source                     df         SS         MS        F  p-value
```

```
──────────────────────────────────────────────────────────────────────
recovery_routine             1  4736.4268  4736.4268  214.370  < 0.001
pony_type                    2  2040.8336  1020.4168   46.184  < 0.001
recovery_routine:pony_type   2  1248.4229   624.2114   28.252  < 0.001
Residuals                   84  1855.9480    22.0946
──────────────────────────────────────────────────────────────────────


Hypotheses (recovery_routine:pony_type):
H₀: There is no interaction between recovery_routine and pony_type
H₁: There is an interaction between recovery_routine and pony_type
Test statistic: F(2, 84) = 28.252
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

2c. Conduct the appropriate hypothesis test to show that there is an interaction effect.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $F =$
- Degrees of freedom:
- $p =$

Rejection Region

- Reject H_0 if ...

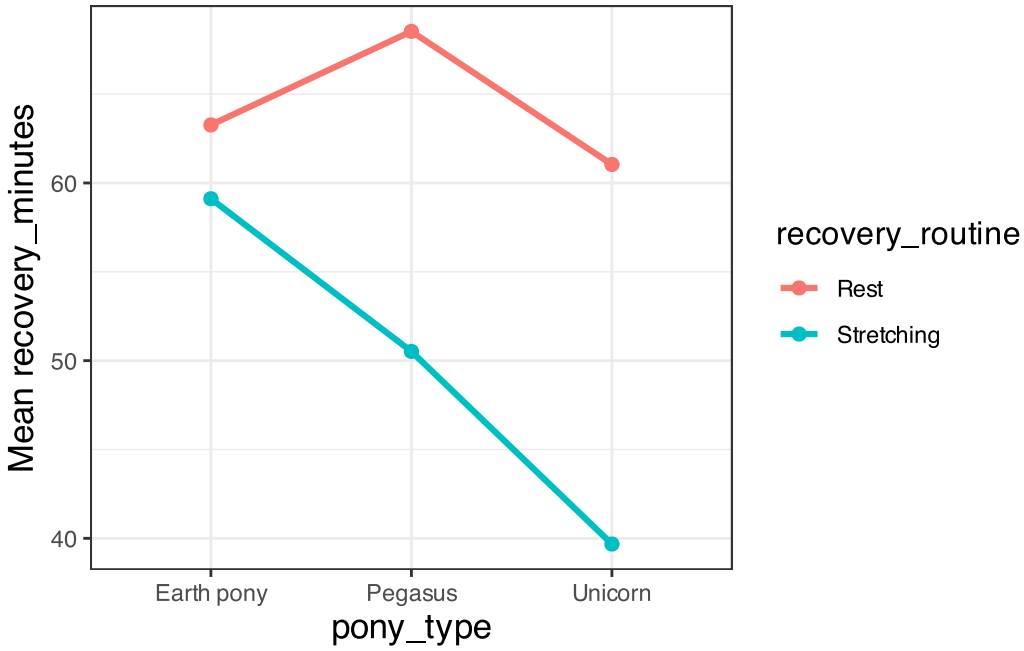
Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

2d. Construct the appropriate posthoc test for a confirmatory study.

2e. Construct a profile plot for this data.

```
applebuck_recovery |> profile_plot(outcome = recovery_minutes,
                                   xaxis = pony_type,
                                   lines = recovery_routine)
```



1f. Write a few sentences to describe what's going on with the data using the results of the ANOVA, the posthoc testing, and the profile plot above.

Your answer here.